

Route Alignment: runA $\rightarrow$ runB

Phase-2 take-home — technical report

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1 Characterising the recordings

runA/runB are two recordings of the same route (cameras **cam0**, **cam5**), delivered as raw HEVC elementary streams (no container – almost no reliable per-file metadata) plus a sidecar **.timestamps.txt** per camera (one capture timestamp per line, ns since epoch, loaded as **int64** throughout – a raw epoch value exceeds **float64**’s exact-integer range). Declared fields came from **ffprobe/ffmpeg** header parsing; measured values came from forcing a full decode and from timestamp-file arithmetic. Code: `src/routealign/{io_video,timestamps}.py`; full JSON per file: `outputs/inspection/`.

Run/cam	Resolution	Size (MB)	Frames (dec./lines)	Decl. fps	Meas. fps	Route (s)	Recorded (UTC)
runA/cam0	1440×1080	129.0	2674 / 2695	10.00	9.992	267.5	2025-02-28 06:24
runA/cam5	1440×1080	121.9	2674 / 2695	10.00	9.992	267.5	2025-02-28 06:24
runB/cam0	1440×1080	70.0	2622 / 2663	10.00	9.989	262.4	2024-04-13 09:12
runB/cam5	1440×1080	64.0	2615 / 2663	10.00	9.989	261.7	2024-04-13 09:12

Table 1: Task 1 summary (full precision in `outputs/inspection/*.json`).

runB was recorded 321 days before **runA** – two separate real-world sessions. A first-frame check supports this: **runA** is sunny with hard shadows; **runB** is overcast with water droplets visible on the camera housing (both cameras), consistent with rain – a qualitative read from individual frames, not a measured statistic.

Frame rate. Three candidate declarations were found and tested, not taken at face value. *Kept*, *10.00 fps*: the HEVC SPS/VUI (`ffmpeg -bsf:v trace_headers`) gives `time_scale/num_units_in_tick=10/1`, independently matching `ffprobe`’s `r_frame_rate`. The **measured** rate, 9.992/9.989 fps, comes from none of these: it is $(N_{\text{decoded}} - 1)/\text{span}$, span from the timestamp file. It is close to the declared value, but that closeness is a finding, not an assumption.

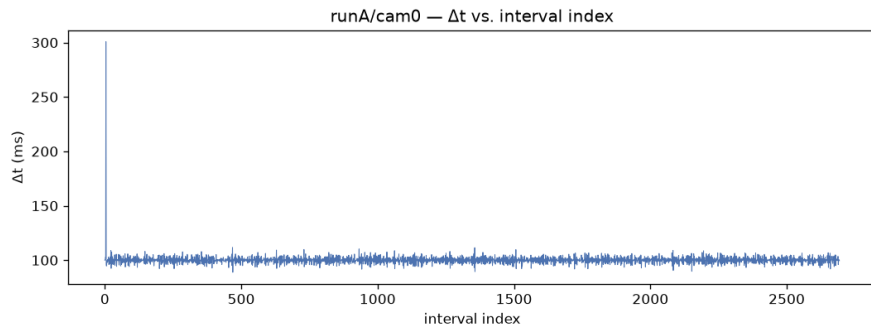


Figure 1: Interval between consecutive frames vs. index, **runA/cam0** (all four files show the same shape). A single 300.8 ms spike (runA) / 396.9 ms (runB) sits in the first few frames; the two frames bracketing each gap are visually indistinguishable, ruling out a scene event – most likely a one-off recording start-up cost, mechanism unconfirmed. No other file shows any comparable gap.

Frame-count discrepancy. Every file has more timestamp lines than frames that actually decode (runA: 2695 vs. 2674, both cameras; runB: 2663 vs. 2622(cam0)/2615(cam5) – *not* identical between cameras, so the usable range must be tracked per camera). All four files are an exact multiple of 262144 B (256 KiB), consistent with – not proof of – a writer that stopped without a final flush; no truncation warning appears on decode either way, and no byte-level NAL scan has confirmed an

incomplete final unit. Consequence for Task 2: the mapping’s H0 convention (timestamp line k = decoded frame k) holds only up to each file’s decoded count; lines beyond it have no video and are marked as such, never forced into a match.

2 Matching frames between runA and runB

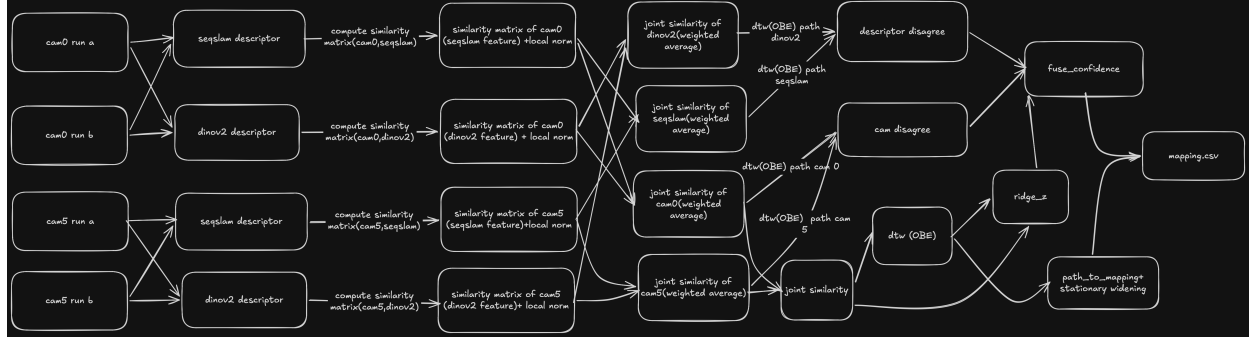


Figure 2: Method pipeline, both runs $\times$ both cameras $\times$ both descriptors through to `mapping.csv`. Four similarity matrices (camera $\times$ descriptor) fuse two ways – per camera and per descriptor – before combining into one $\hat{S}_{\text{joint}}$; `dtw_open_ends` (“OBE”) runs on the joint matrix and, separately, on each single-axis matrix, so the single-axis paths can disagree with the joint one (`cam_disagree`, `desc_disagree`); `ridge_z` from the joint path and the two disagreement cues feed `fuse_confidence`.

2.1 Method

Four decisions, per the brief. **Representation**: two descriptors, fused rather than chosen between – a classical SeqSLAM-style patch-normalised grayscale descriptor (cheap, illumination-robust) and a pretrained DINOv2 ViT-S/14 embedding (semantic, learned), each computed per camera, per frame. **Search**: cosine similarity, then *local* contrast normalisation along each runA row (window tuned per descriptor – 50 for SeqSLAM, 500 for DINOv2, §4), fused into one $\hat{S}_{\text{joint}}$ by a z-scored weighted average across both cameras and both descriptors. **Constraint**: both runs traverse the route in the same order, so the match must be monotone – dynamic time warping with open ends (`dtw_open_ends`), never an unconstrained per-row argmax. **Confident vs. wrong**: four agreement cues computed once the path is fixed – `ridge_z` (path strength along its own trajectory), `margin_z` (claimed match vs. the best alternative outside a ± 30 -column band), `cam_disagree` (cam0-only vs. cam5-only path), `desc_disagree` (SeqSLAM-only vs. DINOv2-only path) – fused into an ordinal 4-tier confidence plus an explicit no-match rule (`runB_frame` left empty). Concretely: **0.9** requires `ridge_z` ≥ 1.5 with both disagreements ≤ 3 frames; **0.7** requires `ridge_z` ≥ 1.0 with both disagreements ≤ 5 ; **0.5** marks rows whose match is a stationary plateau (a range, not one frame); everything else that survives stays at **0.4** – matched, but clearing no tier’s bar. No-match fires on `ridge_z` $< \tau = 0.75$ (τ from a synthetic deletion test, not a guess), on a boundary-clamped row with `ridge_z` < 1.5 , or on both disagreements > 30 . The tiers are an ordinal ranking whose meaning is *measured* (the reliability table, Table 2), not a calibrated probability.

Alternatives considered: a single descriptor was rejected – SeqSLAM alone is cheap but collapses under rain/low-texture stretches, DINOv2 alone has no classical baseline to cross-check against; early fusion (concatenating descriptors before similarity) was tested for real (§2.3), not just argued about.

2.2 Results

`outputs/mapping.csv`: 2695 rows, one per runA timestamp line. **status**: matched 2488, ambiguous_range 124 (inside a stop/plateau – real match, not frame-exact), no_match 62 (abstained), no_frame 21 (H0 tail, no decoded frame at all). Confidence tiers: 0.9 (926 rows), 0.7 (1104), 0.5

(24), 0.4 (558).

Confidence tier	n (scored anchors)	correct	measured accuracy
0.9	8	8	100%
0.7	7	5	71%
0.4	4	2	50%

Table 2: Reliability table: measured accuracy per confidence tier, against 19 user-verified anchors. Monotonic – confidence is an honest ordinal ranking, not just a number.

2.3 Ablation

Configuration	position error (20 anchors)			pred. at no-GT		ridge_z-only gate		
	median	p90	hit@5	@2600	@2650	f.-abst.	z@2600	z@2650
argmax on S_{joint} (no DTW)	0.0	556.3	0.65	655	655	0/20	1.59	1.47
SeqSLAM only, late fusion	0.0	79.2	0.85	2597	2600	1/20	0.36	1.17
SeqSLAM only, early fusion	0.0	3.7	0.90	2598	2600	0/20	0.90	1.46
DINOV2 only, late fusion	0.0	4.3	0.90	2606	2614	1/20	0.56	0.78
DINOV2 only, early fusion	0.0	4.1	0.90	2604	2614	0/20	0.46	0.79
All-4 early fusion (concat)	0.0	3.5	0.90	2597	2600	0/20	0.39	1.02
cam0 only (both descriptors)	0.0	3.7	0.90	2600	2600	0/20	0.75	1.31
cam5 only (both descriptors)	2.0	7.9	0.85	2585	2587	2/20	0.94	0.90
Joint late fusion – shipped	0.0	3.5	0.90	2597	2600	1/20	0.36	0.79

Table 3: Ablation, one row per configuration (early/late fusion = descriptor concatenation vs. score averaging). *Left*: position error in frames against the 20 of 22 anchors that have a ground-truth position – the other 2 (2600, 2650) are genuine loop-closure no-matches with no position to score (§2.4). *Middle*: the raw DTW candidate at those 2 anchors – a bare path cannot output “no match”, it always returns some column. *Right*: a **ridge_z**-only no-match gate, same $\tau = 0.75$ /window 5 as the shipped pipeline but without **cam_disagree**/**desc_disagree** (most configurations have no second, independently-computed path to disagree with); f.-abst. = how many of the 20 real-position anchors it would wrongly abstain on. Bold ridge = caught ($< \tau$); bold elsewhere = the notable value in its column.

Four findings worth defending individually. **DTW is doing real work**: p90 556.3 $\rightarrow$ 3.5 frames once monotonicity is added – the quantitative version of Fig. 3’s corner aliasing. **cam5 is measurably worse**: median 2.0 vs. 0.0, p90 7.9 vs. 3.7 – cam5 (road-facing) sees more passing traffic than cam0 (kerb-facing), confirmed visually in Fig. 4. **Early vs. late camera fusion, tested not argued**: SeqSLAM benefits hugely from early fusion (79.2 $\rightarrow$ 3.7 – late fusion’s score averaging lets one camera’s false attractor dominate; concatenation needs *joint* agreement first); DINOV2 barely changes (4.3 $\rightarrow$ 4.1). Shipped late-fusion still matches early fusion’s best result (3.5 vs. 3.7) through a different mechanism (descriptor, not camera, redundancy) while keeping **cam_disagree**/**desc_disagree** computable, which full early fusion would preclude. **Recognising “no match” is hard for every configuration**: at the 2 no-GT anchors all 8 DTW rows agree on *where* to within 29 frames of each other (argmax’s 655/655 is Fig. 3’s corner aliasing again – two rows 50 apart collapsing onto one wrong attractor column); the ridge-only gate catches 2650 for none of the 9 and 2600 for five. The shipped **fuse_confidence** correctly suppresses 2600 (**ridge_z**=0.36) but lets 2650 through at confidence 0.4 (**ridge_z**=0.79 $\geq \tau$), and its one false abstain (anchor 183, **ridge_z**=0.69) is identical with or without the disagreement cues, cross-checked against **outputs/evaluation_anchors.csv** – on this sample those cues changed no abstain decision; their contribution is the reliability table’s tier separation, not the no-match rule.

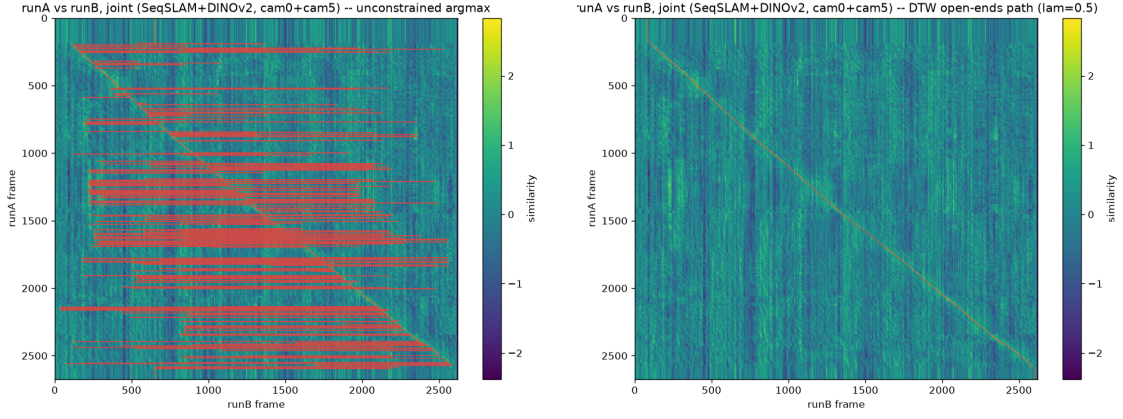


Figure 3: Unconstrained per-row argmax (left) aliases onto visually similar but wrong columns; DTW with the monotonicity constraint (right) tracks the true diagonal.

2.4 Ground truth and an honest accuracy estimate

22 stratum-S anchors (runA 200–2500 roughly every 115 frames, plus the plateau at 60/150 and the loop-closure tail at 2600/2650), labelled by a human from model-blind browsing sheets (**gt-sheets**) against a fixed protocol (**gt/labelling_protocol.md**: same place = within ± 0.5 s of runB travel, lane offset ignored) – never used for tuning. 19 of 22 have a position to score (3 are **no_match** on one side or the other, see below). **median error 0.0 frames, p90 4.2 frames, hit@2/5/10 89.5%**, Wilson 95% CI on hit@5 [68.6%, 97.1%] – quoted because $n = 19$ is small enough that the interval, not just the point estimate, is the honest number to defend. A dev/test split (6/16, as originally planned) was considered and **deliberately not done**: splitting 22 further leaves too few anchors on either side for a usable interval, and the deletion test (recall 1.0, precision 0.56 on a synthetic runB deletion) already covers what the split was for – threshold sanity, not a second accuracy number.

The 3 unscored anchors are informative, not discarded. Anchor 183: ground truth [102, 104], pipeline’s own raw candidate (column 100) missed by only 2 frames but abstained anyway – **ridge_z**=0.69 fell 0.06 short of $\tau = 0.75$. Anchor 2600: correctly abstained, **ridge_z**=0.36, **cam_disagree**=15.0 (both cameras genuinely disagree). Anchor 2650: **ridge_z**=0.79 passed τ but ground truth is a genuine loop-closure **no_match** – a wrong guess let through just above the line. Net read: $\tau = 0.75$ is not free of cost in either direction at this sample size, and 3 points is descriptive, not a basis for retuning it.

2.5 Failure cases

One anticipated failure mode was checked for and **not found**: "runA tail beyond runB’s truncated end" (**boundary_clamped**). Zero rows have it set in the shipped mapping – the two runs’ route coverage happens to reach both open ends. Reporting this honestly is worth more than forcing an example that does not occur.

Where the mapping is least reliable: the depot/carport area (anchors 60, 150 – a visually-repetitive plateau where many runA frames are near-identical), the loop-closure tail (2600–2673, genuinely ambiguous by construction), and any stretch under dense tree canopy (weak margin, not wrong per se, but low-confidence by design). **cam5-only** performance is measurably worse than **cam0-only** throughout, consistent with its greater exposure to transient occlusion.

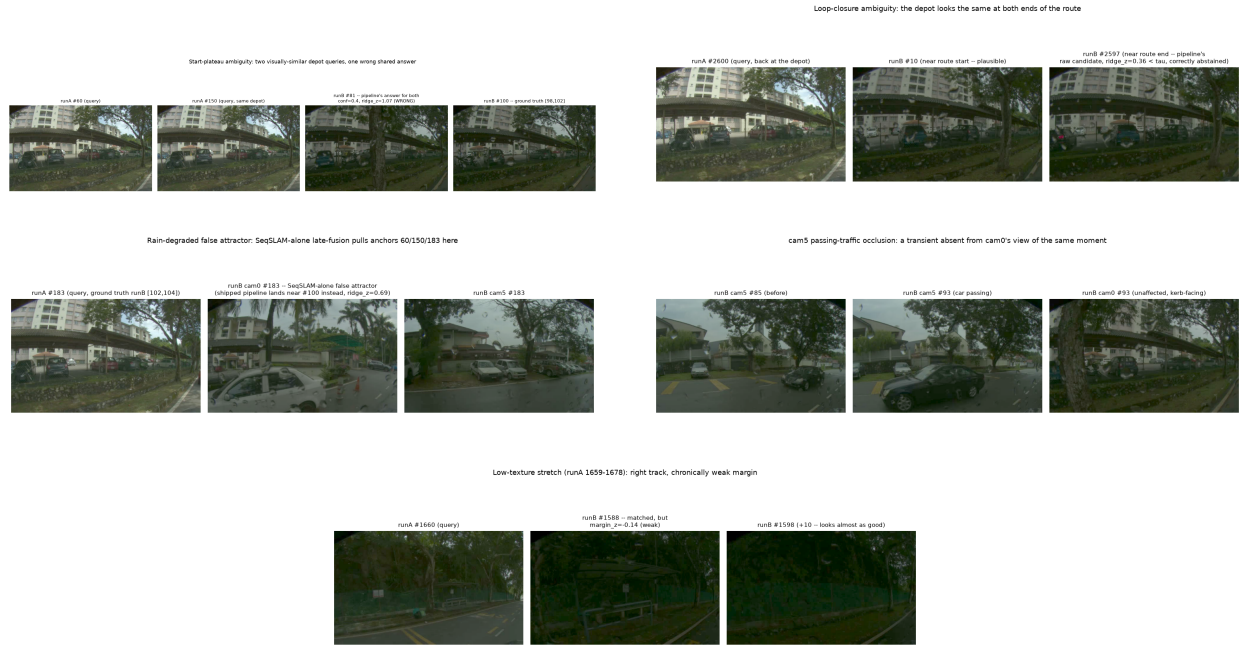


Figure 4: Five real failure modes. *Top left*: identical depot frames (runA 60, 150) collapse onto the same confident-wrong answer. *Top right*: loop closure – runB #10 and #2597 are nearly the same photo, so abstaining is correct. *Middle left*: the rain-degraded scene (runB #183) SeqSLAM’s late fusion collapses onto for three anchors. *Middle right*: a passing car occludes cam5 (motion energy peaks then drops) but not cam0 at the same index. *Bottom*: dense canopy (runA 1659–1678) tracks correctly but with chronically negative `margin_z` – weak discrimination, not one wrong attractor.

3 Keep/discard policy

`outputs/keep_manifest.csv`: one row per (run, cam, frame) – not per timestamp line – because decode status, occlusion and motion genuinely differ per camera even at the same index. $2695 \times 2 + 2663 \times 2 = 10716$ rows. **keep=10585, discard=131** – 131 exactly reproduces the independently-measured `no_frame` gap (21+21+41+48), a real cross-check, not a coincidence.

Only one rule is a hard discard: no decoded frame. Every other rule attaches a column a downstream consumer filters on explicitly, deliberately **flag-heavy rather than discard-heavy** – a stationary frame is redundant for training diversity but might be exactly what a "does the model recognise a parked scene" eval wants; discarding it pre-emptively answers a question we were not asked. Attached per surviving frame: timing confidence (`ts_confidence`, tail-zone flag), interval regularity, stationary-run membership and a 1 Hz dedup flag, run/camera-level weather, lighting and occlusion tags, Task 2’s correspondence (`corr_*`, "not_applicable" for runB rows – a reverse mapping was never computed or evaluated, so it is not invented here), and a `place_id` for split-leakage-safe train/val/test assignment (contiguous 70/15/15 blocks over $\lfloor \text{runB_frame}/100 \rfloor$, both runs’ frames of one place always sharing a split, the shared start/end depot merged into one id and left **unassigned**).

Corrupt decode and exact duplicates – checked, not assumed: `motion_energy` read exactly 0.0 for zero consecutive-frame pairs across all 4 streams; re-decoding all 4 streams at a stricter ffmpeg verbosity than production (`-v warning` vs. `-v error`) produced zero warning lines. **keep=decode_ok** is evidence-backed as complete for this dataset, not merely the only rule that happened to get built.

4 Reproducibility and the proved-vs-assumed ledger

Full reproduction steps, pinned/measured dependency versions and the `ffmpeg` build string are in `README.md`; every number in this report was read directly from the pipeline’s own regeneratable outputs (`outputs/mapping.csv`, `outputs/keep_manifest.csv`, `outputs/evaluation_summary.json`, `notebooks/results/`), not transcribed from memory. `make all` twice produces byte-identical CSVs (fixed seeds, sorted iteration, no wall-clock values in any output).

Claim	Status	Evidence
H0 (frame k = ts. line k = decoded frame k)	assumed	GOP/IDR positions match exactly across all 4 streams for their whole usable range; no hardware timestamp cross-check exists
Declared 10 / measured 9.99 fps	proved	VUI matches <code>ffprobe</code> ; measured is $(N_{\text{decoded}} - 1) / \text{span}$ from timestamps – independent sources
Reliability table monotonicity	measured	$n = 4/7/8$ per tier – real, not cherry-picked, but too small to detect a mild mis-calibration
No corrupt/duplicate frames	proved	zero warnings at stricter verbosity; zero exact-zero <code>motion_energy</code> transitions
<code>depot_runb_margin</code> , 70/15/15	split	chosen a defensible convention, not the one correct policy
Weather / lighting / <code>lens_occlusion</code>	assumed	one-time visual read per run/camera, not a per-frame detector

Table 4: Every claim in this report that is not a direct, repeatable measurement.